Lecture 6 - The Complex Exponential

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October 8, 2025

- The *steady state* of a system is the long term behaviour of a system's outputs.

1 Compute the outputs to the Finite Impulse Response (FIR) system $y(n) = x(n) + x(n-2)$ for the following inputs:

1.a $x(n) = \cos(\frac{1}{2}\pi n)$

n	0	1	2	3	4	5
$x(n)$	1	0	-1	0	1	0
$y(n)$	1	0	0	0	0	0

Here, we can observe that the system's steady state is 0.

1.b $x(n) = \cos(\pi n)$

n	0	1	2	3	4	5
$x(n)$	1	-1	1	-1	1	-1
$y(n)$	1	-1	2	-2	2	-2

Here we can observe that the system's steady state is ± 2.

Notice here that despite the difference equations being the same, $y(n)$'s steady state state differs depending on the frequency of the input cosine.

- Trying to evaluate what the steady-state behaviour of a system is by using the difference equation formula is largely fruitless.
  - Expressing our input and output in terms of complex exponentials, is the key to understanding a system's steady state.
- The complex exponential form for sine and cosine are given respectively as:

$$\sin(\omega n) = \frac{1}{2}(e^{i\omega n} + e^{-i\omega n})$$
$$\cos(\omega n) = \frac{1}{2i}(e^{i\omega n} - e^{-i\omega n})$$

2 Compute the outputs to the system $y(n) = x(n) + x(n-2)$for the input$x(n) = e^{i\omega n}$

$$\begin{aligned} y(n) &= x(n) + x(n-2) \\ &= e^{i\omega n} + e^{i\omega(n-2)} \\ &= e^{i\omega n} + e^{i\omega n - 2i\omega} \\ &= e^{i\omega n} + e^{i\omega n}e^{-2i\omega} \\ &= (1 + e^{-2i\omega})e^{i\omega n} \end{aligned}$$

- From example 2, we can see that our output $y(n)$ is equal to our input $x(n) = e^{i\omega n}$ times some complex constant.
  - This is generally true for all systems.
  - This complex constant is called the system's *frequency response*.
    * Denoted $H(\omega)$
- In the same way that a system's impulse response, $h(n)$, describes a system's output when its input is the Kronecker-Delta function $x(n) = \delta(n)$, $H(\omega)$ describes a system's output when its input is a complex exponential $x(n) = e^{i\omega n}$.
- $H(\omega)$ describes how the system scales and phase-shifts complex exponential inputs.

3 Evaluate the outputs to the system $y(n) = x(n) + x(n-2)$ for the following values of ω:

3.a $\omega = \frac{1}{2}\pi$

$$\begin{aligned} y(n) &= (1 + e^{-2i\omega})e^{i\omega n} \\ &= (1 + e^{-2i(\frac{1}{2}\pi)})e^{i(\frac{1}{2}\pi)n} \\ &= (1 + e^{-\pi i})e^{\frac{1}{2}\pi i n} \\ &= (1 - 1)e^{\frac{1}{2}\pi i n} \\ &= 0 \end{aligned}$$

3.b $\omega = \pi$

$$\begin{aligned} y(n) &= (1 + e^{-2i\omega})e^{i\omega n} \\ &= (1 + e^{-2i(\pi)})e^{i(\pi)n} \\ &= (1 + e^{-2\pi i})e^{\pi i n} \\ &= (1 + 1)e^{\pi i n} \\ &= 2(-1)^n \end{aligned}$$

Note that the outputs of these systems for complex exponentials whose frequencies corresspond to the angular frequencies of the cosines given in example 1 also correspond to their steady-state behaviour!

Complex Numbers Recap

- i is defined to be $\sqrt{-1}$
  - A complex number is any number z which can be expressed as $z = a + bi$, where $a, b \in \mathbb{R}$.
    - $\mathbb{R}$ denotes the set of all real numbers.
    - $\mathbb{C}$ denotes the set of all complex numbers.
    - a is called the real component of z, $Re(z) = a$.
    - b is called the imaginary component of z, $Im(z) = b$.
    - The complex conjugate of z is defined as $a - bi$
      - Denoted $\bar{z}$
    - the argument of z is the angle formed between the vector pointing from the origin to z and the positive real axis in the complex plane.
      - Denoted $arg(z)$
      - For signals, the sign of $arg(z)$ is associated with being either forward or backwards in time.
    - The modulus of z is the length of the vector pointing from the origin to z.
      - Denoted $|z|$
      - Given by $|z| = \sqrt{a^2 + b^2}$
- Some useful theorems:
  - $Re(z) = \frac{1}{2}(z + \bar{z})$
  - $Im(z) = \frac{1}{2}(z - \bar{z})$
- All exponential numbers, $z \in \mathbb{C}$ can be written in *exponential form*.
  - A complex number is in exponential form if it is some multiple of e raised to some complex power.
  - Given by $z = |z|e^{iarg(z)}$.

4 Express the following complex numbers in exponential form (with a negative argument)

4.a $z = -i$

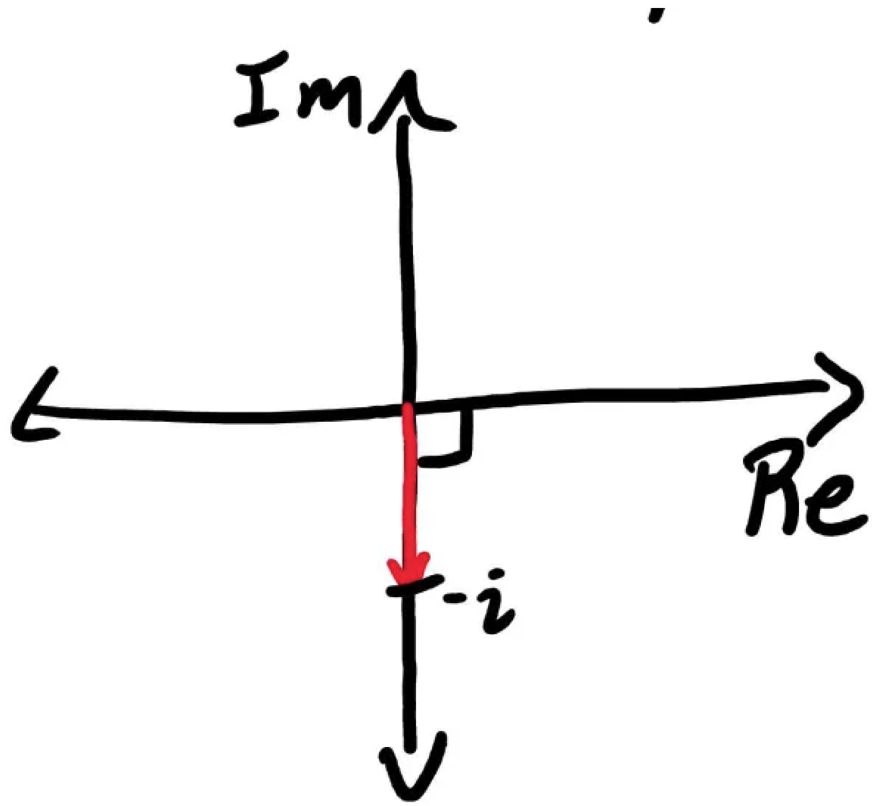

Figure 1: z in the complex plane

$$
\begin{aligned}
|z| &= \sqrt{(a^2 + b^2)} \\
&= \sqrt{(0^2 + (-1)^2)} \\
&= \sqrt{1} \\
&= 1 \\
arg(z) &= \frac{-\pi}{2}
\end{aligned}
$$

Thus, $z = -i = e^{\frac{-\pi}{2}i}$

4.b $z = 1 + i$

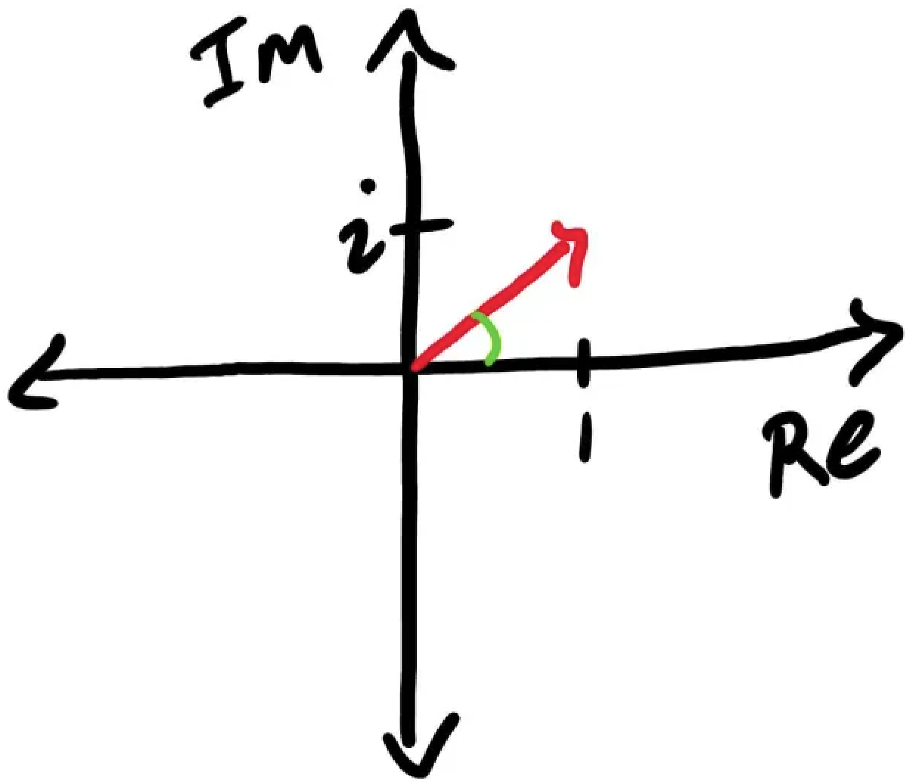

Figure 2: z in the complex plane

$$\begin{aligned} |z| &= \sqrt{(a^2 + b^2)} \\ &= \sqrt{(1^2 + 1^2)} \\ &= \sqrt{2} \\ arg(z) &= \frac{-7\pi}{4} \end{aligned}$$

Thus $z = 1 + i = \sqrt{2}e^{\frac{-7\pi}{4}i}$